



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering

Academic Year 2025 – 2026 (Even Semester)

Degree, Semester & Branch: II Semester B.E. CSE 'B'

Course Code & Title: CS25C07 Object Oriented Programming

Name of the Faculty member (s): Dr.B.Vijayalakshmi, ASP-II/CSE

Innovative Practice Description

- **Unit / Topic: Module 5 / Templates and Exceptions**

- **Course Outcome: CO5**

- **Activity Chosen: Flipped classroom**

- **Justification:**

The flipped classroom approach encourages students to learn basic concepts before attending the classroom session, thereby improving active participation and conceptual understanding during class hours. For the topic “Strings”, students were provided with learning resources such as video lectures, PPT materials, and reference notes before the class. This enabled student to familiarize themselves with string declaration, initialization, string manipulation functions, and operations in advance. The classroom session was then utilized for collaborative discussions, problem-solving activities, and clarification of doubts, resulting in improved engagement and deeper understanding of string concepts in Object-Oriented Programming.

- **Time Allotted for the Activity: 45 Minutes**

- **Details of the Implementation:**

- Before the classroom session, students were instructed to study the learning materials shared through canvas consists of reference materials and notes related to strings in C++.
- The pre-class materials covered topics such as declaration and initialization of strings, string input/output operations, concatenation, comparison, string handling functions, and member functions of string class.
- Students were asked to prepare short notes and identify doubts while studying the materials independently.
- During the classroom session, the faculty member conducted a quick recap discussion to assess students' understanding of the pre-learned concepts.
- Students were then grouped into teams of 6 members for collaborative activities involving string-based programming exercises and debugging tasks as shown in Fig 1 & 2.
- String functions utilization in different types of problem such as checking palindrome strings, reversing strings, counting vowels, and string comparison were discussed and implemented.

- Students actively participated in peer discussions and clarified conceptual doubts through interaction with faculty members and classmates.
- The session concluded with a short quiz and discussion to reinforce the concepts learned.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO5	3	2	3	1	2	-	-	1	1	1	1	3	3	3

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO9	PO10	PSO1
	3	2	1	1	3
Justification for correlation	The students gain knowledge of string handling concepts and operations in C++.	The students analyze and solve string manipulation problems effectively.	The students work collaboratively during classroom activities and peer discussions.	The students improve communication skills by presenting solutions and clarifying doubts.	String handling concepts are essential for developing efficient object-oriented applications.

• **Images / Screenshot of the practice:**

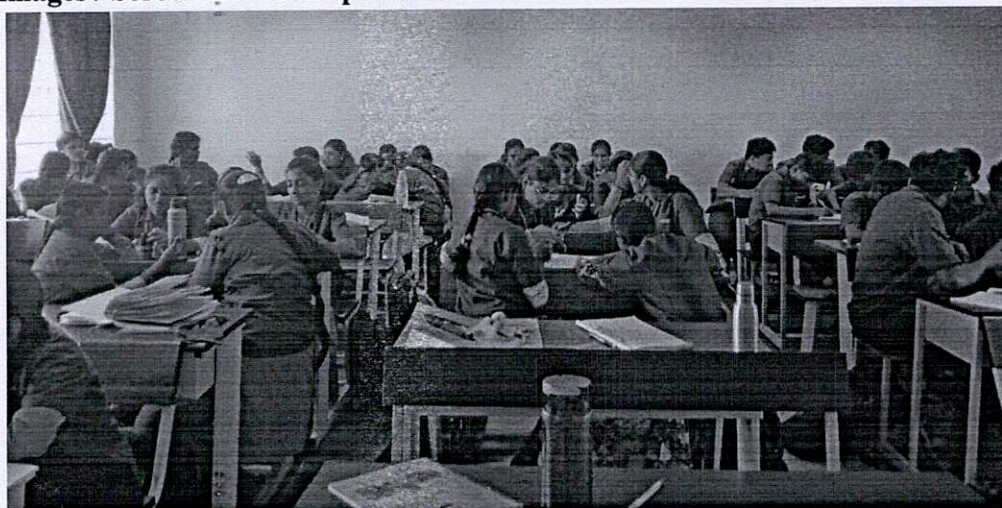


Fig 1. Student participation during flipped classroom discussion

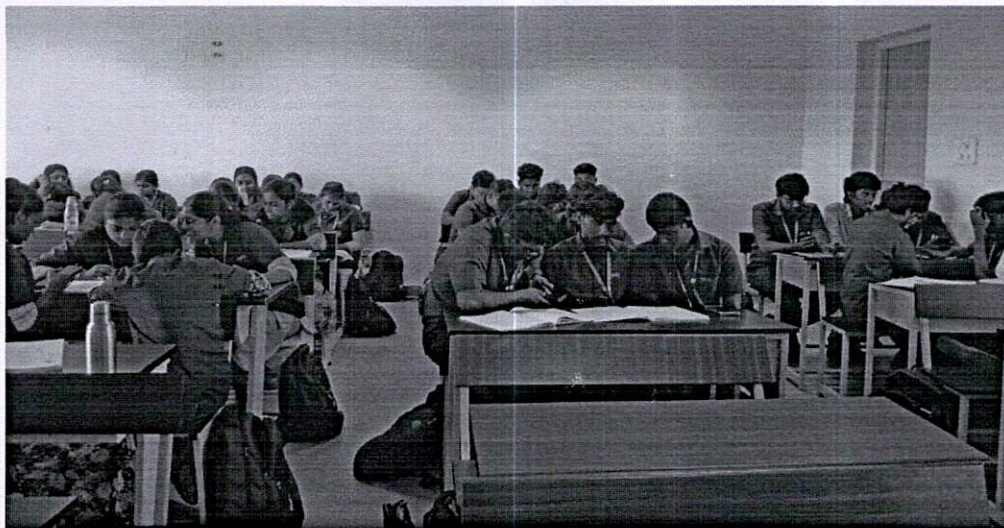


Fig 2. Students solving string manipulation programs collaboratively

- **Reflective Critique:**

- ❖ *Feedback of practice from students and other stakeholders:*

- Students reported that learning the concepts before class helped them participate more actively during classroom discussions.
 - The activity improved confidence in solving string-related programming problems.

- ❖ *Benefit of the practice:*

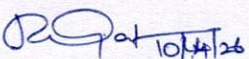
- The flipped classroom model improved student engagement and promoted self-learning.
 - Students demonstrated better understanding and retention of string handling concepts through collaborative activities and practical problem solving.

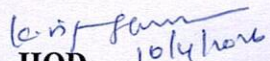
- ❖ *Challenges faced in implementation:*

- Some students initially faced difficulty in completing the pre-class preparation.
 - Additional guidance and follow-up discussions were required for slow learners to actively participate in the activities.

References:

- <https://uwaterloo.ca/centre-for-teaching-excellence/catalogs/tip-sheets/course-design-planning-flipped-class>
- https://www.ritrjpm.ac.in/images/computer-science/2024-2025/IP/Unit_4_Flipped%20Classroom.pdf


Signature of Faculty Member


HOD